

Managing Moisture



Allen Zimmerman is a professor of engineering technology and technical physics at the Ohio State University, Wooster Campus.

Q I enjoyed reading your article on the home you recently built in Ohio (“Design, Construction, and Performance in Ohio,” Jan/Feb ’07, p. 36). I am an architect practicing in the Pacific Northwest (Lopez Island, Washington). At our firm we have been designing the rain screen systems for a few years with great response and success. Currently we are exploring the insulated sheathing aspect of the rain screen with different cladding options. I know we live in two completely different climate zones and this may affect the details of the construction process. That being said, I have a few questions about your recent project:

1. Did you utilize any other moisture barrier over the OSB other than the 2-inch rigid foam? I am also curious what you used for the cladding/siding on your house and if it required air ventilation.
2. Did you create an air space between the exterior cladding and the 2-inch rigid foam? Here in the Northwest we utilize this air space in order to keep the siding material (wood, most commonly) as dry as possible on both sides. We usually accomplish this by using pressure-treated 1-inch strapping over the sheathing.

3. With the thicker wall system in your Ohio home, did you experience any problems with window installation? Did you utilize a nailing fin-type window or a jamb-mounted window system? I assume that the nail fin was attached at the exterior face of the rigid foam (drainage plane), if this was the window type used. Following the recommendations of Joe Lstiburek of Building Science Corporation (and now most of the moisture barrier manufacturers), flashing the window around the nailing fin to the drainage plane is the best way to go. I’m curious how you detailed this while building your house.

Todd Kegerreis
Greene Partners
Lopez Island, Washington



The exterior cladding of this home is vinyl siding, which reduces moisture problems, is durable, low-maintenance, and low cost.



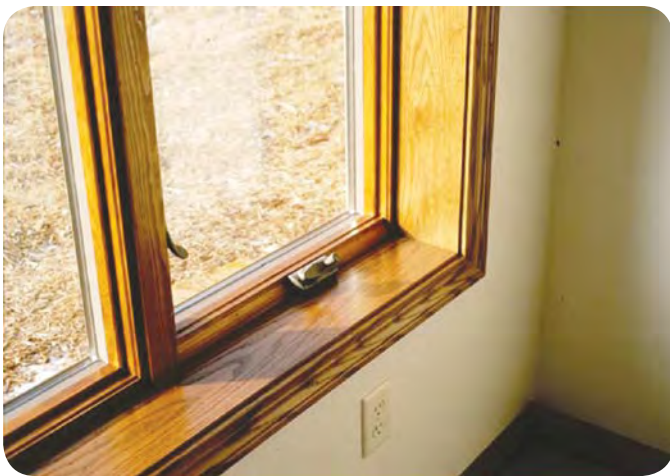
These windows are attached to 2 x 2 strips of wood placed around the rough opening in the wall. Self-adhesive membrane strips function as the window flashing, with the sill piece applied prior to window installation and the sides and top pieces applied after window installation.

A. The exterior cladding of my home in Ohio is vinyl siding. As you know, vinyl siding is back vented by design, structure, and installation, and this simplifies the exterior-wall detail. Therefore, I was able to (1) have the 2-inch rigid foam (taped at the joints) function as the drainage plane, and (2) apply the siding directly to the foam. Given the 2-inch thickness of the foam, I had the siding subcontractor use 3-inch nails spaced 12 inches on center (siding manufacturers typically recommend 16-inch spacing) and placed so that every other nail penetrated through the OSB into a stud (exterior wall studs are spaced 24 inches on center). The

lack of moisture-related problems is a major advantage of vinyl siding. Other practical advantages include durability, low maintenance, ease and simplicity of installation, and low cost.

It is sometimes said that vinyl siding is not green. Betsy Pettit (of Building Science Corporation) did an excellent job of addressing this topic in her response to a reader who asked this very question in the June 2005 issue of *Fine Homebuilding* magazine (p. 102) a couple of years ago. Pettit made the following points:

- All materials have environmental costs associated with their use. In her experience as a building performance consultant, problems with wood siding not being installed correctly and therefore failing prematurely are common.



Here are the interior trim details for windows in the 8 1/2-inch-wide exterior walls.

- She has never investigated a moisture- or mold-related problem with vinyl siding.
- The use of recycled content in manufacturing vinyl siding, and recycling at the job site and at replacement, are all becoming more common.

After discussing in detail the real-world pros and cons of vinyl siding versus wood and other materials, Pettit concludes that, yes, vinyl can be green, and I agree.

The windows that I used on my house had nailing fins. I was not comfortable mounting the nailing fins directly to the 2-inch rigid foam. Therefore, windows are attached to 2 X 2 strips of wood placed around the rough openings in picture frame fashion. Self-adhesive membrane strips function as the window flashing, with the sill piece applied prior to window installation and the sides and top pieces applied after window installation. This technique for window installation and flashing proved to be very effective and problem-free.

